

Adult Census Income Prediction using Machine Learning

Project Overview

This project focuses on predicting whether an individual's annual income exceeds **\$50K** using the Adult Census Income Dataset. The project includes data preprocessing, exploratory data analysis (EDA), machine learning model development, ensemble learning, hyperparameter tuning, and structured JSON-based model analysis.

Dataset

- **Dataset:** Adult Census Income Dataset
 - **Target Variable:** `income`
 - **Task:** Binary Classification
 - **Language:** Python
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Technologies Used

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Matplotlib
 - Scikit-learn
 - Joblib
 - JSON
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Project Structure

Part 1: Data Preprocessing & Exploratory Data Analysis

- Data Loading
- Data Cleaning
- Missing Value Analysis
- Statistical Summary
- Data Visualization
- Correlation Analysis

Part 2: Machine Learning Models

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Model Evaluation
- Confusion Matrix
- Classification Report
- Feature Importance
- Cross Validation
- Pipeline Creation

Part 3: Advanced Machine Learning

- Voting Classifier
- Bagging Classifier
- AdaBoost Classifier
- Gradient Boosting Classifier
- Hyperparameter Tuning using GridSearchCV
- Ensemble Model Comparison
- Best Model Selection

Part 4: AI-powered Structured Analysis

- Model Summary
- JSON Output Generation
- Structured Insights
- Model Interpretation
- Final Performance Analysis

Best Performing Model

Random Forest Classifier

This model achieved the best performance and was selected as the final model for income prediction.

Files Included

- Adult_Census_Income.ipynb
- adult.csv
- best_model.pkl
- best_ensemble_model.pkl
- model_summary.json
- structured_insights.json
- model_comparison.csv

- ensemble_results.csv
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Results

- Data preprocessing completed successfully.
 - Multiple machine learning models were trained and evaluated.
 - Random Forest achieved the best overall performance.
 - Ensemble learning and hyperparameter tuning improved model performance.
 - Structured JSON outputs were generated for model analysis.
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Author

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